

AI Impact in Nepal's Industries and Sectors

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I. ABSTRACT

The impact of Artificial Intelligence (AI) on different industries and sectors in Nepal is examined in this study report. We investigate the use of AI in agriculture, healthcare, finance and education through a thorough analysis of the literature that has already been published. The results demonstrate AI's ability to change established procedures, boost output, and raise the quality of services. Opportunities for AI-driven growth and innovation amplify, despite issues like data privacy and job replacement. This study emphasizes the necessity of adopting AI strategically and implementing supportive policies in order to maximize its advantages for Nepal's sustainable development.

II. PURPOSE

This report's goal is to investigate, assess, and predict the probable future effects of Artificial Intelligence (AI) on different sectors and businesses in Nepal. We want to know how AI innovations are presently changing conventional techniques and improving productivity, thus we are conducting a thorough examination. Additionally, by identifying the opportunities and difficulties that lie ahead, this paper aims to predict the future role of AI in revolutionizing various sectors of Nepal.

III. BACKGROUND

Between 1950 and 1956, the emergence of artificial intelligence (AI) showed a crucial turning point. The Turing Test was developed to determine computer intelligence as a result of Alan Turing's work on "Computer Machinery and Intelligence". During this time, the phrase "artificial intelligence" was first used, and Arthur Samuel's checker-playing program became the first to have the ability to learn on its own.[5].

The 1980s and 1990s saw tremendous developments in the field of artificial intelligence [5].

IV. INTRODUCTION

Artificial intelligence (AI) is the ability of digital computers or robots to carry out actions that are typically performed by intelligent beings. The creation of systems with cognitive abilities similar to those of humans, such as reasoning, meaning discovery, generalization, and learning from past experiences, is the aim of AI development [6]. The technological revolution is being led by artificial intelligence (AI), which is changing industries, economics, and societies all around the world.

There are many types of AI, but Weak AI and Strong AI stand out as the two most common [7].

- A. Weak AI: Sometimes referred to as Narrow AI, describes AI systems that are specialized and created to carry out particular tasks within predetermined circumstances [7]. These systems lack general human-like cognitive capacities and only display intelligence in the context of their assigned tasks.

Virtual assistants, image recognition softwares, and recommendation systems are a few examples of weak AI. These systems excel in their respective fields but lack full consciousness or intelligence.

- B. Strong A: It is also known as General AI or Artificial General Intelligence (AGI) [7]. Strong AI attempts to create robots with intellect similar to that of humans, with the capacity to understand, pick up new information, and reason about a variety of topics. Strong AI is still a work in progress because it requires creating systems with creative and self-awareness capabilities.

V. AI LANDSCAPE IN NEPAL

At the moment, over 96% of Nepalese homes with at least one mobile phone also have smartphones [1]. Siri on iPhones and Google Assistant on Android devices serve as examples of Nepal's limited AI access.

Innovators in Nepal's AI and machine learning fields include startups like Fusemachines Nepal and Paaila Technology. "Pari", a robot from Paaila Technology that is currently operating for restaurants in Kathmandu and has facial recognition and customer identification capabilities, was revealed [2]. In Kathmandu, some restaurants have begun deploying these robot waiters, and the owner has noticed a favorable impact on the company. The company has also created Nepali-language voice recognition software. Although this is good progress, it is not enough to keep up with global events.



Figure 1. Pari Robot from Paaila Technology

On the other hand, the data gathered at the time of the 2015 Earthquake was used to trace the mass movement, which was made possible by large-scale data analysis [2]. Worldwide, there have been updates in the AI feature like ChatGPT, DALL-E and mid journey. Nepal as a part of global civilization is also touched by these changes.

VI. AI APPLICATION

Artificial intelligence (AI) has the power to alter a number of industries in Nepal. Here are some applications of AI that Nepal may use:

A. Agricultural Sector:

AI can have an immediate impact in farming [3]. Smart agriculture is possible with the use of AI. It can help in predicting the best harvesting time. Farmers can use pesticides on only the affected plants with the help of image data processed by AI technology.

B. Tourism Growth:

Through the use of chatbots, AI can be used to offer tourists personalized service and information. Additionally, tourists can have immersive explorations of Nepal's cultural and natural attractions through the use of virtual reality (VR) and augmented reality (AR) technologies [9].

C. Microfinance organizations:

Similar to successful use by Africa[3], implementing AI in microfinance organizations can analyze massive volumes of data by AI-powered algorithms, which can then detect patterns and trends to determine frauds and reduce financial risks.

D. Government and self-governance:

By automating administrative activities and offering data-driven insights for decision-making, AI can help the government achieve self-governance. This may result in increased governance transparency and efficiency.

E. Employment openings:

The government and enterprises can efficiently shortlist qualified people for employment openings by using Machine Learning (ML) analysis.

VII. CHALLENGES AND OPPORTUNITIES

There are some challenges of AI development in Nepal that needs to be faced:

A. Economic Disruption:

The widespread adoption of AI could potentially lead to a disruption in the economy, causing job losses and financial challenges for many individuals. New AI related job opportunities may emerge, but transitioning from one job to another can be difficult [8].



Figure 2. Effect of AI in Economy

B. Impact on Social and Family:

Reliance on AI and robots to an excessive degree may result in a detachment from social and family connection. Families may split up, birth rates may drop, and poor mental health may spread as a result.

C. Lack of AI Adoption:

Despite having a young population and an abundance of technical expertise, Nepal has lagged behind other nations in adopting AI due to lack of awareness. The absence of AI integration could slow growth and development in a number of different industries.

D. Digital Divide:

Although Nepal is seeing an increase in smartphone use and internet access, there is still a digital divide between urban and rural areas that needs to be resolved in order to promote AI development throughout the nation.

VIII. FUTURE OUTLOOK

In Nepal, the future of AI offers promising possibilities for expansion and growth in a number of industries [4]. Following important recommendations are made for the ethical development and application of AI:

A. Including AI in engineering education:

The existing engineering education needs to be updated in order to promote a generation of AI experts that have a solid idea of ethical issues as well [4]. Nepal can promote this culture by discussing AI related topics with students and encouraging them to think about how their research will affect the actual world.

B. Raising Awareness of AI dangers and Limitations:

It's critical to increase awareness of AI dangers and limitations to society. The general public, professionals, students, and policymakers should all get education on the ethical consequences of AI technologies. Making educated decisions and developing AI responsibly will be made possible by greater awareness.

C. Encouraging Diversity in the AI Community:

Diversity in the AI community is crucial for promoting innovation, inclusion, and a range of viewpoints in the development of AI. In order to promote more comprehensive AI solutions, efforts should be made to improve the participation of women and marginalized groups of people in fields of AI.

IX. CONCLUSION

With institutions and startups like Fusemachines Nepal, AI Expo Nepal and Paaila Technology advancing in the domains of AI and machine learning, Nepal's AI landscape is growing. To keep up with global trends, however, more integration of AI technologies is still required.

In Nepal, artificial intelligence is used in a wide range of fields, including business and government.

The outlook for AI in Nepal is positive going forward. Nepal can set the road for sustainable growth in the upcoming years by inclusion of the government in proper policy making, building like minded communities, organizing and participating in AI events, solving issues, and utilizing the potential of AI. If Nepal gets a good grasp over AI tech, then it can reach up to some extent in overall growth.

X. REFERENCES

- [1] Nepal in the decade of AI, 27 December 2019, from, <https://www.nepalitimes.com/here-now/nepal-in-the-decade-of-ai>, Accessed 5 August 2023.
- [2] Artificial Intelligence in Nepal, 5 August 2023, from, https://english.onlinekhabar.com/pro-artificial-intelligence-in-nepal.html?fbclid=IwAR1KecM2V1dntVIPedi6BnKiCerUL2_PVaNrMkP4wXVF-5U_B9_ziYzQyfQ, Accessed 5 August 2023.
- [3] AI and its Impact in Nepal, 18 August 2022, from, https://www.nepalitelecom.com/2019/01/ai-artificial-intelligence-nepal.html?fbclid=IwAR2i6qQ_8qXSI5AIgqUbBHIZ0QYWi13KaYQJv7w7ByhQX7IT0PK0EMFAXH0, Accessed 5 August 2023.
- [4] AI Ethics Survey in Nepal, 2021, from, https://www.naamii.org.np/wp-content/uploads/2021/11/AI-Ethics-Survey-Report.pdf?fbclid=IwAR1oYLYXGmvrpa_ADC2AlNN-5D09TtV8VGqEiEGqUuZPbDahCUkrpm8VsKs, Accessed 5 August 2023.
- [5] What is the history of Artificial Intelligence?, from, <https://www.tableau.com/data-insights/ai/history>, Accessed 5 August 2023.
- [6] Artificial Intelligence, 4 August 2023, from, <https://www.britannica.com/technology/artificial-intelligence>, Accessed 5 August 2023.
- [7] Types of Artificial Intelligence, 20 December 2018, from, <https://certes.co.uk/types-of-artificial-intelligence-a-detailed-guide/>, Accessed 5 August 2023.
- [8] AI: Its origin, present and scope in Nepal, 2019, from, https://www.youtube.com/watch?v=EilMQ_S1LV0, Accessed 6 August 2023.
- [9] AI and Tourism in Nepal, 1 July 2023, from, <https://isp.page/news/ai-and-tourism-in-nepal-enhancing-visitor-experiences/>, Accessed 6 August 2023